CIE

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TITLE: **Sarthi-RaahSathi**

SCHOOL: Chitkara University, Rajpura, Punjab

DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO SHARE LOGIN AND PASSWORD WITH MENTORS

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Roll No/ Employee Code:

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ABSTRACT

1.1 Problem Background

The logistics and transportation sector faces persistent challenges in shipment allocation, pricing transparency, and regulatory compliance. Drivers often depend on informal networks to secure loads, leading to inconsistent income and limited visibility into fair market pricing. Shippers, on the other hand, struggle to compare competitive bids efficiently and monitor shipment progress in real time. Additionally, expired permits and documentation lapses frequently result in penalties, operational delays, and compliance risks. Existing load-matching platforms focus primarily on connectivity but lack structured bidding mechanisms and integrated permit monitoring systems, creating inefficiencies in the logistics ecosystem.

1.2 Objective of the Project

The objective of Sarthi-RaahSathi is to develop a digital driver–shipper marketplace platform that integrates a transparent bidding system with automated permit notification features to enhance operational efficiency, ensure compliance, and improve decision-making for both drivers and shippers.

1.3 Proposed Solution / System Overview

Sarthi-RaahSathi is a web-based logistics management platform that connects drivers and shippers through a real-time bidding mechanism. Upon authentication, users select their role as either a driver or a shipper. Drivers can access a dashboard displaying active bids, total earnings, completed trips, and ratings, along with detailed shipment listings including image, weight, route, minimum bid, and bidding deadline. Drivers may place and update bids within the permitted timeframe. Shippers can create shipments, compare bids, select drivers, and track shipment progress. The major components of the system include User Authentication Module, Driver Dashboard, Shipper Dashboard, Bidding Management Module, Shipment Tracking Module, and Permit Notification Module.

1.4 Key Features or Functionality

The system offers real-time bid visibility, structured minimum bid control, earnings and trip analytics, shipment lifecycle tracking, and driver rating mechanisms. A dedicated permit management module allows drivers to add permit details and receive automated alerts before expiration. The bidding timer ensures fair competition, while dashboard summaries provide quick operational insights for both drivers and shippers.

1.5 Technology / Method Used

The platform is developed using Vite and React for frontend development, Clerk for secure authentication management, and Supabase for backend services and database handling. The system follows a client-server architecture model to ensure scalability, data integrity, and efficient role-based access control.

1.6 Expected Outcome & Benefits

Sarthi-RaahSathi aims to create a transparent and compliance-driven logistics ecosystem. The platform is expected to increase income visibility for drivers, enable better bidding decisions for shippers, reduce permit-related legal risks, and streamline shipment management processes. By integrating bidding and compliance monitoring in a single system, operational efficiency and trust between stakeholders are significantly improved.

1.7 Original Contribution Statement

Sarthi-RaahSathi introduces a unified logistics marketplace combining real-time bidding transparency with automated permit compliance monitoring, offering an integrated operational and regulatory solution within a single digital platform.

SYSTEM DESIGN

2.1 Architecture

The system follows a client-server architecture where the frontend communicates with backend services through APIs.

2.2 Modules

- Authentication Module
- Driver Dashboard
- Shipper Dashboard
- Bidding Management
- Shipment Tracking
- Permit Management

2.3 Data Flow

Level 0: User interacts with system → processes → database

Level 1: Authentication → Dashboard → Bidding → Shipment → Permit

2.4 Use Case

Actors:

- Driver
- Shipper

Use cases:

- Login/Register
- Place Bid
- Create Shipment
- Track Shipment
- Manage Permits

2.5 ER (Entity Relationship)

Entities:

- User
- Shipment
- Bid
- Permit Relationships define interactions among them.

IMPLEMENTATION

3.1 Technologies Used

- Frontend: React (Vite)
- Backend: Supabase
- Authentication: Clerk

3.2 Functionalities

- Real-time bidding system
- Shipment listing and tracking
- Dashboard analytics
- Permit expiry notifications

3.3 Workflow

1. User logs in
2. Selects role
3. Interacts with dashboard
4. Performs actions (bidding/shipment)

RESULTS AND DISCUSSION

The system successfully:

- Improved transparency in pricing
- Reduced manual coordination
- Enhanced compliance tracking
- Increased operational efficiency

Graphs and performance metrics can be added for better analysis.

CONCLUSION

The Sarthi-RaahSathi platform successfully addresses key challenges in the logistics and transportation sector by introducing a unified digital marketplace for drivers and shippers. By integrating a transparent real-time bidding system with automated permit compliance monitoring, the platform reduces dependency on informal networks, enhances pricing visibility, and ensures regulatory adherence.

The system's modular architecture and use of modern technologies such as React, Supabase, and Clerk enable scalability, secure authentication, and efficient data handling. Features like dashboard analytics, shipment tracking, and permit notifications contribute to improved operational control and informed decision-making for all stakeholders.

Overall, Sarthi-RaahSathi enhances trust, minimizes inefficiencies, and promotes a more structured and reliable logistics ecosystem. The platform lays a strong foundation for future enhancements such as AI-driven insights, GPS tracking, and mobile accessibility, making it a scalable and forward-looking solution in the logistics domain.

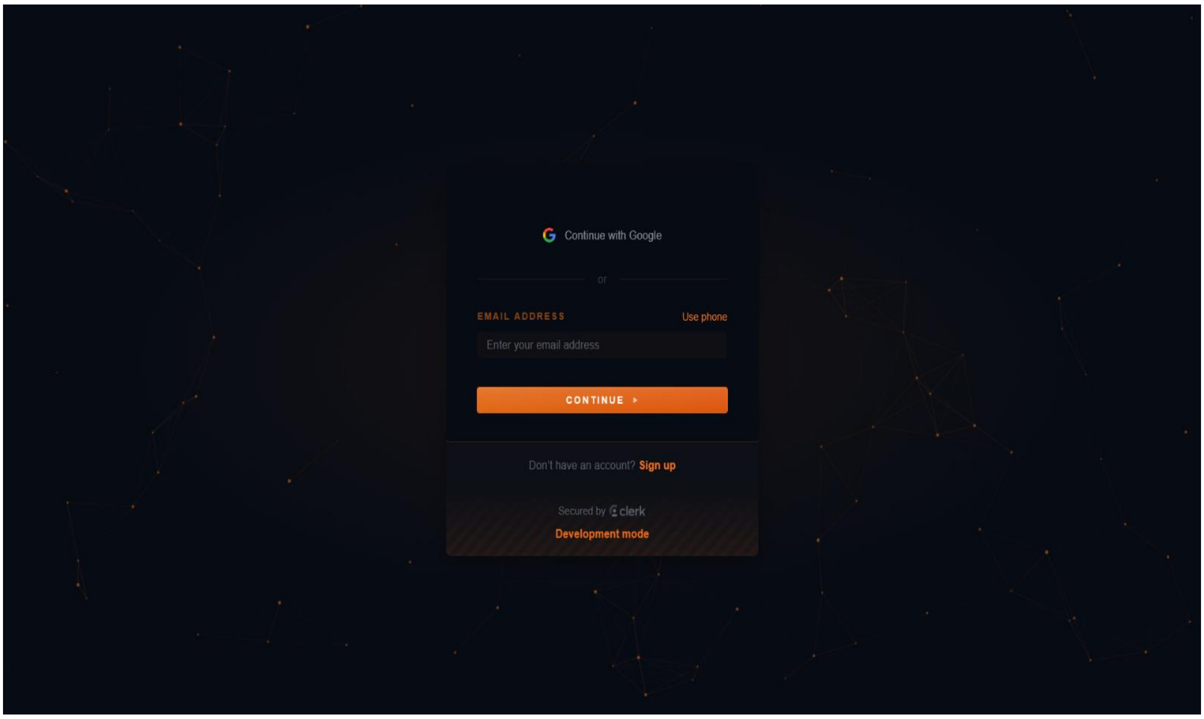
FUTURE SCOPE

- AI-based bidding suggestions
- Real-time GPS tracking
- Mobile application development
- Payment gateway integration

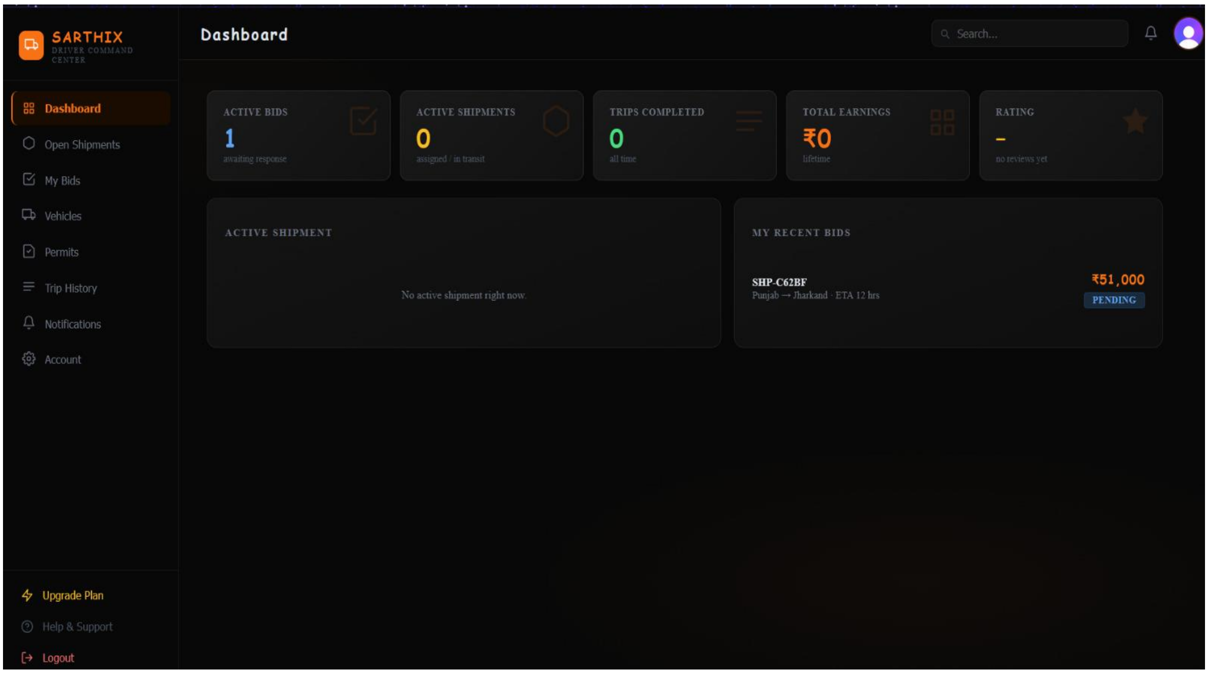
APPENDIX

Screenshots

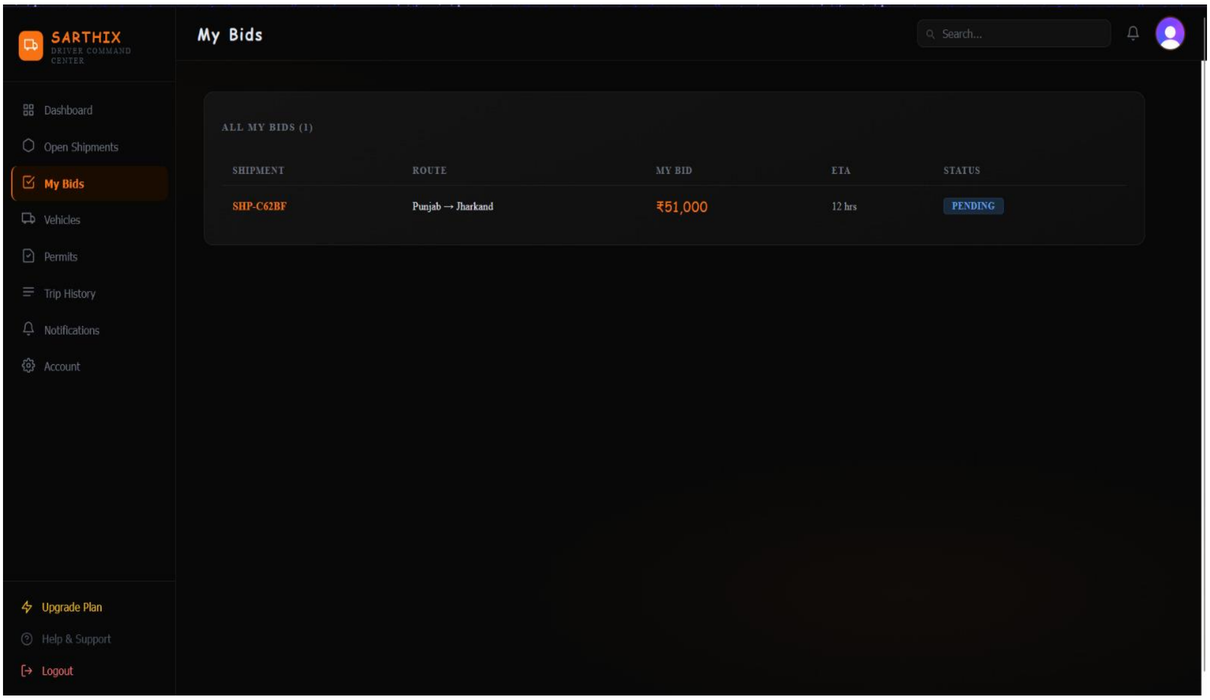
- Figure 1: Sign in / Sign up



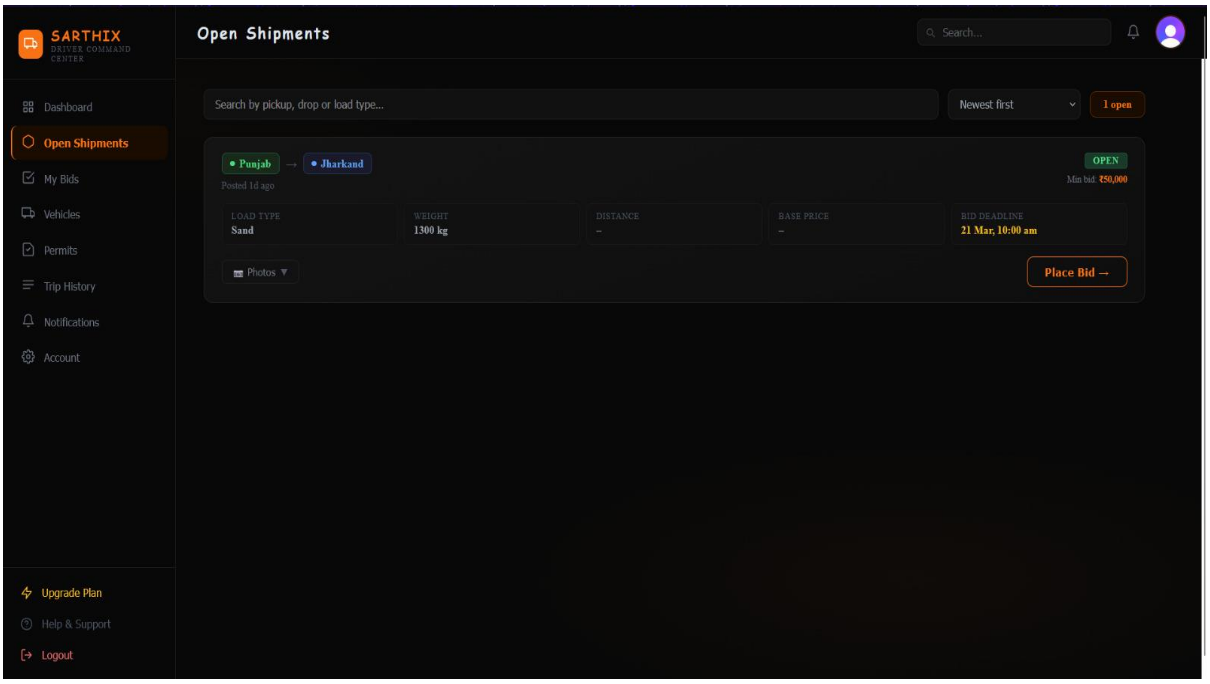
- Figure 2: Dashboard



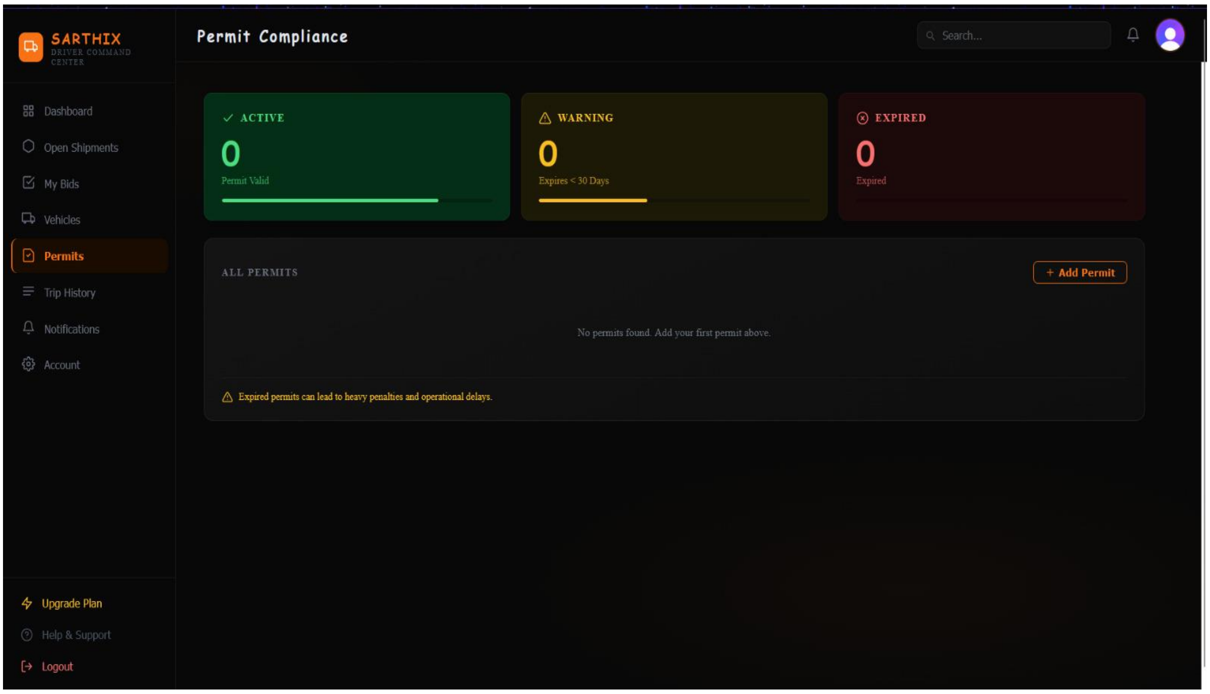
- Figure 3: Bidding Page



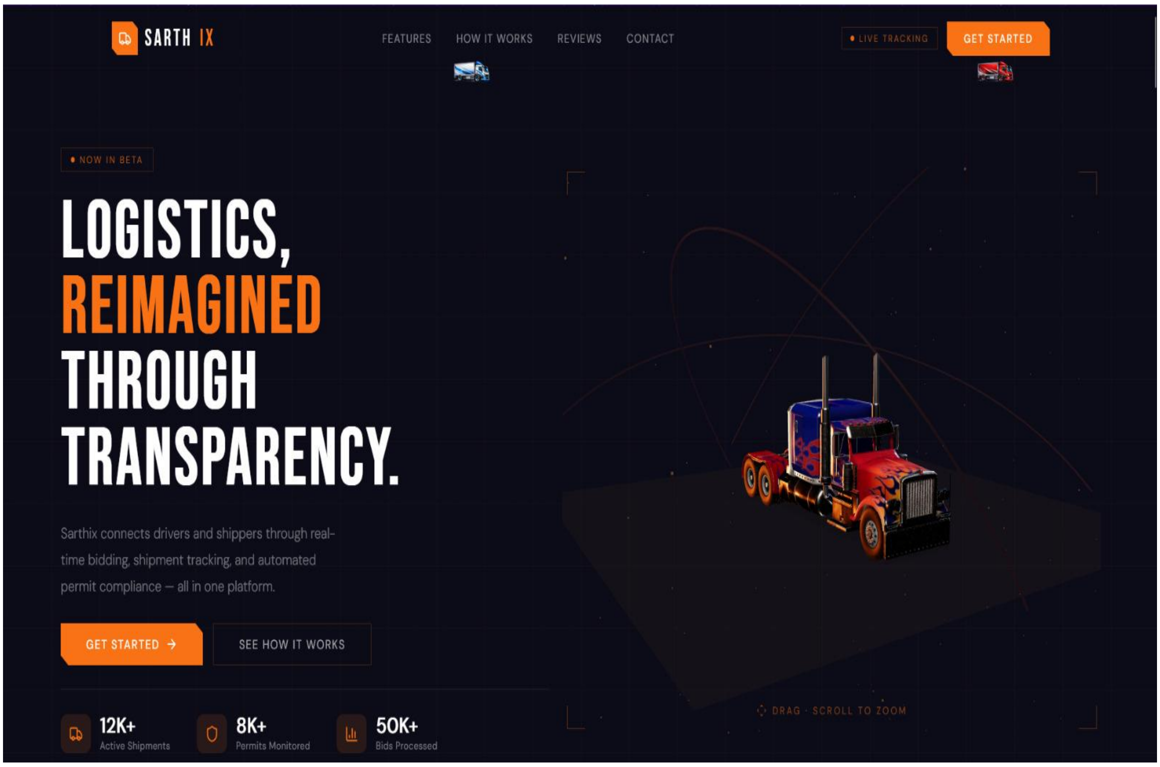
- Figure 4: Shipment Page



- Figure 5: Permit Page

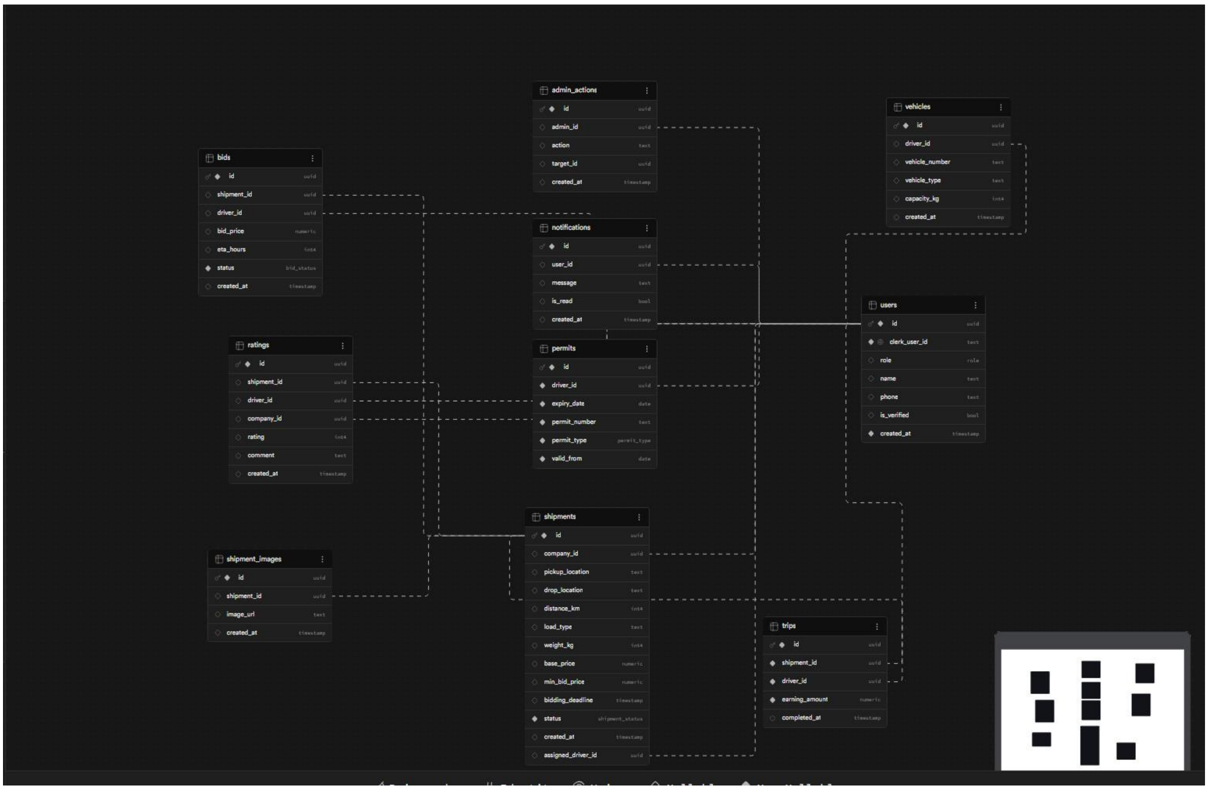


- User Actions



- Diagrams

Data Flow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phone text
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